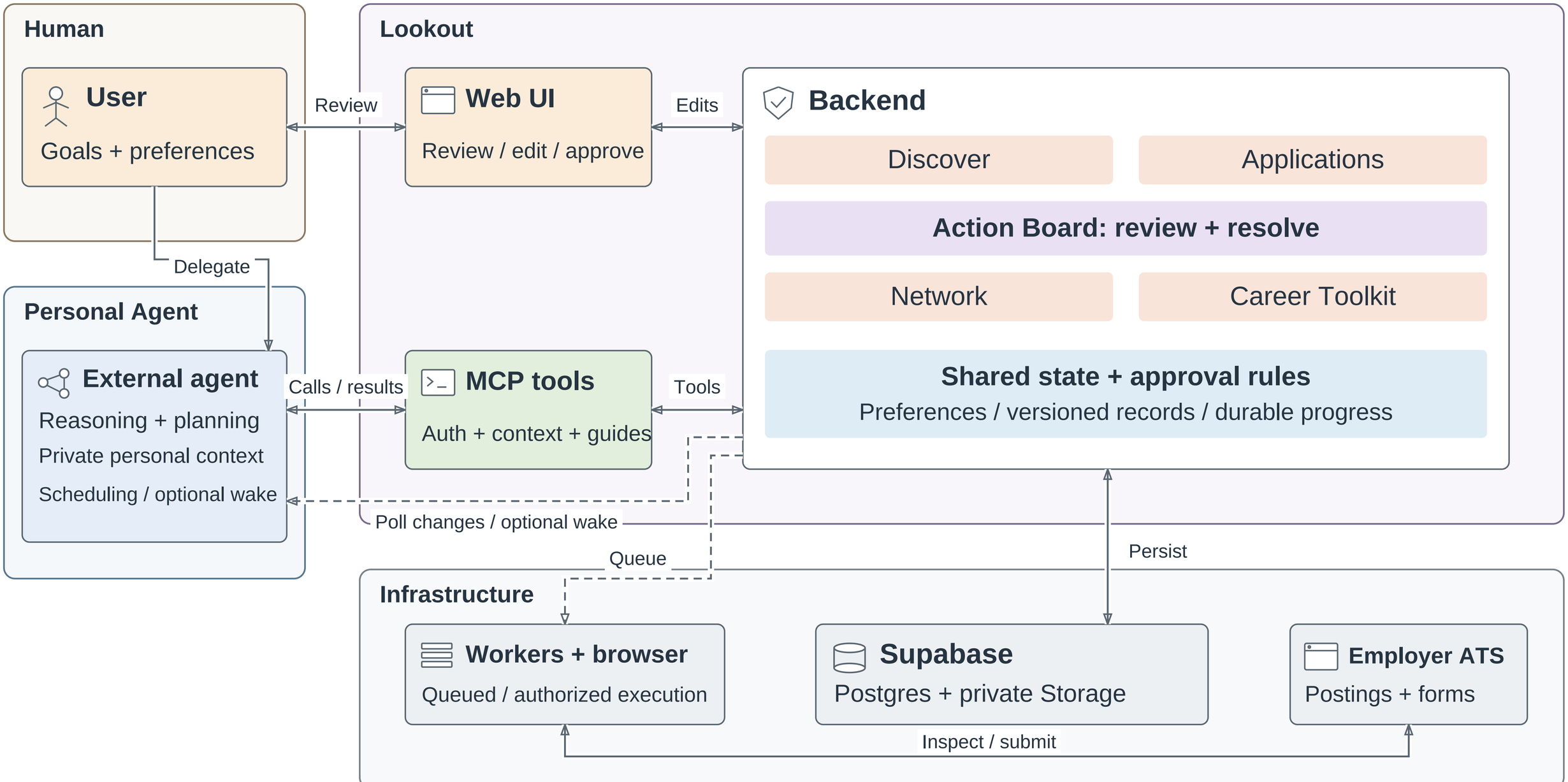Lookout architecture